



Intel® Core™ Ultra 5 Processor 125H

18M Cache, up to 4.50 GHz



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Essentials

|                                                                                                              |                                                          |
|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Product Collection                                                                                           | <a href="#">Intel® Core™ Ultra processors (Series 1)</a> |
| Code Name                                                                                                    | <a href="#">Products formerly Meteor Lake</a>            |
| Vertical Segment                                                                                             | Mobile                                                   |
| Processor Number            | 125H                                                     |
| Overall Peak TOPS (Int8)  | 33                                                       |

CPU Specifications

|                                                                                                                                  |                          |
|----------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Total Cores                                   | 14                       |
| # of Performance-cores                                                                                                           | 4                        |
| # of Efficient-cores                                                                                                             | 8                        |
| # of Low Power Efficient-cores                                                                                                   | 2                        |
| Total Threads                                 | 18                       |
| Max Turbo Frequency                           | 4.5 GHz                  |
| Performance-core Max Turbo Frequency          | 4.5 GHz                  |
| Efficient-core Max Turbo Frequency            | 3.6 GHz                  |
| Low Power Efficient-core Max Turbo Frequency  | 2.5 GHz                  |
| Performance-core Base Frequency               | 1.2 GHz                  |
| Efficient-core Base Frequency                 | 700 MHz                  |
| Low Power Efficient-core Base Frequency       | 700 MHz                  |
| Cache                                         | 18 MB Intel® Smart Cache |
| Processor Base Power                          | 28 W                     |
| Maximum Turbo Power                           | 115 W                    |
| Minimum Assured Power                                                                                                            | 20 W                     |

Feedback

|                                                                       |                               |
|-----------------------------------------------------------------------|-------------------------------|
| Intel® Deep Learning Boost (Intel® DL Boost) on CPU <a href="#">?</a> | Yes                           |
| AI Software Frameworks Supported by CPU                               | OpenVINO™, WindowsML, ONNX RT |
| CPU Lithography <a href="#">?</a>                                     | Intel 4                       |

#### Supplemental Information

|                                                  |                          |
|--------------------------------------------------|--------------------------|
| Marketing Status                                 | Launched                 |
| Launch Date <a href="#">?</a>                    | Q4'23                    |
| Embedded Options Available <a href="#">?</a>     | Yes                      |
| Use Conditions <a href="#">?</a>                 | PC/Client/Tablet         |
| Product Tuning (Embedded Uses) <a href="#">?</a> | Yes                      |
| Datasheet                                        | <a href="#">View now</a> |

#### Memory Specifications

|                                                              |                                                  |
|--------------------------------------------------------------|--------------------------------------------------|
| Max Memory Size (dependent on memory type) <a href="#">?</a> | 96 GB                                            |
| Memory Types <a href="#">?</a>                               | Up to LPDDR5/X 7467 MT/s<br>Up to DDR5 5600 MT/s |
| Max # of Memory Channels <a href="#">?</a>                   | 2                                                |
| ECC Memory Supported † <a href="#">?</a>                     | No                                               |

#### GPU Specifications

|                                                                 |                                                                         |
|-----------------------------------------------------------------|-------------------------------------------------------------------------|
| GPU Name† <a href="#">?</a>                                     | Intel® Arc™ graphics                                                    |
| Graphics Max Dynamic Frequency <a href="#">?</a>                | 2.2 GHz                                                                 |
| GPU Peak TOPS (Int8) <a href="#">?</a>                          | 18                                                                      |
| Graphics Output <a href="#">?</a>                               | eDP1.4b, DP 2.1 UHBR20, HDMI 2.1 FRL                                    |
| X <sup>e</sup> -cores                                           | 7                                                                       |
| Max Resolution (HDMI)† <a href="#">?</a>                        | 4096 x 2304 @ 60Hz (HDMI 2.1 TMDS)<br>7680 x 4320 @ 60Hz (HDMI 2.1 FRL) |
| Max Resolution (DP)‡ <a href="#">?</a>                          | 7680 x 4320 @ 60Hz                                                      |
| Max Resolution (eDP - Integrated Flat Panel)‡ <a href="#">?</a> | 3840 x 2400 @ 120Hz                                                     |
| DirectX* Support <a href="#">?</a>                              | 12.2                                                                    |
| OpenGL* Support <a href="#">?</a>                               | 4.6                                                                     |
| OpenCL* Support <a href="#">?</a>                               | 3.0                                                                     |
| H.264 Hardware Encode/Decode <a href="#">?</a>                  | Yes                                                                     |
| H.265 (HEVC) Hardware Encode/Decode <a href="#">?</a>           | Yes                                                                     |
| AV1 Encode/Decode <a href="#">?</a>                             | Yes                                                                     |
| Intel® Quick Sync Video <a href="#">?</a>                       | Yes                                                                     |
| # of Displays Supported †                                       | 4                                                                       |

|                                                                       |                                                 |
|-----------------------------------------------------------------------|-------------------------------------------------|
| Device ID                                                             | 0x7D55                                          |
| Intel® Deep Learning Boost (Intel® DL Boost) on GPU <a href="#">?</a> | Yes                                             |
| AI Software Frameworks Supported by GPU                               | OpenVINO™, WindowsML, DirectML, ONNX RT, WebGPU |
| NPU Specifications                                                    |                                                 |
| NPU Name <sup>‡</sup>                                                 | Intel® AI Boost                                 |
| NPU Peak TOPS (Int8) <a href="#">?</a>                                | 11                                              |
| Sparsity Support                                                      | Yes                                             |
| Windows Studio Effects Support                                        | Yes                                             |
| AI Software Frameworks Supported by NPU                               | OpenVINO™, WindowsML, DirectML, ONNX RT         |
| Expansion Options                                                     |                                                 |
| Intel® Thunderbolt™ 4 <a href="#">?</a>                               | Yes                                             |
| PCI Express Revision <a href="#">?</a>                                | 5.0 and 4.0                                     |
| PCI Express Configurations <sup>‡</sup> <a href="#">?</a>             | 1x8 5.0 + 3x4 4.0 + 2x4 (x1,x2,x4) 4.0          |
| Max # of PCI Express Lanes <a href="#">?</a>                          | 28                                              |
| Package Specifications                                                |                                                 |
| Sockets Supported <a href="#">?</a>                                   | FCBGA2049                                       |
| Max Operating Temperature <a href="#">?</a>                           | 110 °C                                          |
| Package Size                                                          | 50 mm x 25 mm                                   |
| Advanced Technologies                                                 |                                                 |
| Intel® Gaussian & Neural Accelerator <a href="#">?</a>                | 3.5                                             |
| Intel® Thread Director <a href="#">?</a>                              | Yes                                             |
| Intel® Smart Sound Technology <a href="#">?</a>                       | Yes                                             |
| Intel® Wake on Voice <a href="#">?</a>                                | Yes                                             |
| Intel® High Definition Audio <a href="#">?</a>                        | Yes                                             |
| Intel® Adaptix™ Technology <a href="#">?</a>                          | Yes                                             |
| Intel® Speed Shift Technology <a href="#">?</a>                       | Yes                                             |
| Intel® Turbo Boost Max Technology 3.0 <sup>‡</sup> <a href="#">?</a>  | Yes                                             |
| Intel® Turbo Boost Technology <sup>‡</sup> <a href="#">?</a>          | 2.0                                             |
| Intel® Hyper-Threading Technology <sup>‡</sup> <a href="#">?</a>      | Yes                                             |
| Intel® 64 <sup>‡</sup> <a href="#">?</a>                              | Yes                                             |
| Instruction Set <a href="#">?</a>                                     | 64-bit                                          |
| Instruction Set Extensions <a href="#">?</a>                          | Intel® SSE4.1, Intel® SSE4.2, Intel® AVX2       |

|                                         |     |
|-----------------------------------------|-----|
| Enhanced Intel SpeedStep® Technology ⓘ  | Yes |
| Thermal Monitoring Technologies ⓘ       | Yes |
| Intel® Volume Management Device (VMD) ⓘ | Yes |

## Security & Reliability

|                                                                     |                        |
|---------------------------------------------------------------------|------------------------|
| Intel vPro® Eligibility ‡ ⓘ                                         | Intel vPro® Essentials |
| Intel® Threat Detection Technology (TDT)                            | Yes                    |
| Intel® Active Management Technology (AMT) ‡ ⓘ                       | No                     |
| Intel® Standard Manageability (ISM) ‡ ⓘ                             | Yes                    |
| Intel® Remote Platform Erase (RPE) ‡                                | No                     |
| Intel® One-Click Recovery ‡                                         | No                     |
| Intel® Hardware Shield Eligibility ‡ ⓘ                              | Yes                    |
| Intel® Secure Key ⓘ                                                 | Yes                    |
| Intel® Control-Flow Enforcement Technology ⓘ                        | Yes                    |
| Intel® Total Memory Encryption - Multi Key                          | No                     |
| Intel® AES New Instructions ⓘ                                       | Yes                    |
| Intel® Trusted Execution Technology ‡ ⓘ                             | Yes                    |
| Execute Disable Bit ‡ ⓘ                                             | Yes                    |
| Intel® OS Guard                                                     | Yes                    |
| Intel® Boot Guard ⓘ                                                 | Yes                    |
| Mode-based Execute Control (MBEC) ⓘ                                 | Yes                    |
| Intel® Stable IT Platform Program (SIPP) ⓘ                          | No                     |
| Intel® Virtualization Technology with Redirect Protection (VT-rp) ‡ | No                     |
| Intel® Virtualization Technology (VT-x) ‡ ⓘ                         | Yes                    |
| Intel® Virtualization Technology for Directed I/O (VT-d) ‡ ⓘ        | Yes                    |
| Intel® VT-x with Extended Page Tables (EPT) ‡ ⓘ                     | Yes                    |

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Refer to Datasheet for formal definitions of product properties and features.

‡ This feature may not be available on all computing systems. Please check with the system vendor to determine if your system delivers this feature, or reference the system specifications (motherboard, processor, chipset, power supply, HDD, graphics controller, memory, BIOS, drivers, virtual machine monitor-VMM, platform software, and/or operating system) for feature compatibility. Functionality, performance, and other benefits of this feature may vary depending on system configuration.

Intel processor numbers are not a measure of performance. Processor numbers differentiate features within each processor family, not across different processor families. See <http://www.intel.com/content/www/us/en/processors/processor-numbers.html> for details.

"Announced" SKUs are not yet available. Please refer to the Launch Date for market availability.

Intel® Arc™ graphics only available on select V-Series Intel® Core™ Ultra processor-powered systems with qualifying system thermal design or H-series Intel® Core™ Ultra processor-powered systems with at least 16GB of system memory in a dual-channel configuration. OEM enablement required. Other Intel® Core™ Ultra processor-powered system configurations feature Intel® Graphics. Check with OEM or retailer for system configuration details.

Intel® Iris® Xe Graphics only: to use the Intel® Iris® Xe brand, the system must be populated with 128-bit (dual channel) memory. Otherwise, use the Intel® UHD brand.

Max Turbo Frequency refers to the maximum single-core processor frequency that can be achieved with Intel® Turbo Boost Technology. See <https://www.intel.com/content/www/us/en/gaming/resources/turbo-boost.html> for more information and applicability of this technology.

See <http://www.intel.com/content/www/us/en/architecture-and-technology/hyper-threading/hyper-threading-technology.html?wapkw=hyper+threading> for more information including details on which processors support Intel® HT Technology.

Processors that support 64-bit computing on Intel® architecture require an Intel 64 architecture-enabled BIOS.

Some products can support AES New Instructions with a Processor Configuration update; in particular, i7-2630QM/i7-2635QM, i7-2670QM/i7-2675QM, i5-2430M/i5-2435M, i5-2410M/i5-2415M. Please contact OEM for the BIOS that includes the latest Processor configuration update.

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